

JuneAirways44SafeServer

High-Level Design Document

Airway Merge Safe Server · .NET 8 MVC · Ground & Air Vehicle Classification

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1. Executive Summary

JuneAirways44SafeServer is a production-grade .NET 8 MVC application that manages highway merge-zone infrastructure and — from Phase 4 onward — a full Urban Air Mobility (UAM) traffic management layer. The system ingests vehicle telemetry from four ground sensor source types (physical, satellite, telecom, tracker) and one dedicated air vehicle source type (airflycar / eVTOL), classifies every record deterministically, stores classification results in the database, and renders them in an interactive Three.js 3D scene with perspective, side-elevation, and top-plan views.

The build was delivered across 8 phases, systematically addressing 20 identified issues (C1–C5 security, A1–A8 architecture, D1–D7 design) and adding 6 production enhancements (E1–E6), the altitude/air-scene feature set, a 4-step vehicle classification engine, and complete AirFlyCar simulation with 25 correlated UAM telemetry fields.

Phase	Focus	Key Deliverables	Files
P1	Security Hardening	BCrypt-only auth · ModelState guards · EventType allowlist · IDOR protection	6
P2	Architecture Fixes	Random.Shared · IDOR highway guard · Password masking · env-var priority	5
P3	Design Improvements	Enum enforcement · EF string converters · IVehicleRegistry DI singleton	5
P4	Altitude & Air Scene	AltitudeMeters on 4 models · migration · 3D AirScene view · Three.js	11
P5	Production Enhancements	Rate limiting · Audit trail · /health endpoint · Serilog · Heartbeat	12
P6	Classification Engine	VehicleClassifier · VehicleClass record · VehicleMode/Category columns	8
P7	Scene Completion	Pre-classified events in view · classification result panel · AirScene rewrite	5
P8	AirFlyCar / eVTOL	4 test configs · 25-field generator · vertiport · conflict rings · alt ruler	7
Total		47 files across 8 phases · 3 migrations · 0 open issues	47

2. System Architecture Overview

2.1 Technology Stack

Layer	Technology	Version / Detail
Runtime	.NET / ASP.NET Core MVC	8.0 LTS
ORM	Entity Framework Core	8.0 — SQLite dev / PostgreSQL prod
Authentication	BCrypt session-based	BCrypt.Net-Next 4.0.3 · HttpOnly cookie
3D Rendering	Three.js	r128 CDN · WebGL · shared scene graph
Logging	Serilog	8.0 · Console + rolling file + Seq sink
Rate Limiting	ASP.NET Core RateLimiter	.NET 8 built-in · fixed-window per IP
Database	SQLite / PostgreSQL	EF migrations · MigrateAsync at startup
Caching	IMemoryCache + OutputCache	In-process · 5–10 min TTL · tag eviction
Background	IHostedService	HeartbeatMonitorService · 2 min poll

2.2 Request Pipeline

Browser → Kestrel → Security Headers MW → Serilog Request Logging → Static Files → Routing → RateLimiter → SessionAuthFilter (global, honours [SkipSessionAuth]) → Controller Action → VehicleClassifier (on write) → AppDbContext → SQLite / PostgreSQL → View / JSON Response

2.3 Data Flow — Classification Pipeline

INPUT RECORD (JSON payload) ■ ▼ InputPayloadService.Generate(sourceType, fields) ■ ▼ VehicleClassifier.Classify(payload, sourceType) ■ Step 1b: sourceType=="airflycar" → AIR domain (flight_phase gate) ■ Step 1: altitude_m > 10 → AIR (≤150=air_urban, >150=air_express) ■ Step 2: vehicle_type field → sedan|suv|truck|motorcycle|van ■ Step 3: speed heuristic → ≤25→motorcycle ... >90→van ■ Step 4: confidence score → 0.0–1.0 composite ■ ▼ VehicleEvent {VehicleMode, VehicleCategory, VehicleClassJson} ■ ▼ GET /api/events/live → JSON feed (includes vehicleClassJson) ■ ▼ AirScene.cshtml / Traffic3D/Index.cshtml Panel A: Perspective Panel B: Side/Elevation Panel C: Top/Plan

3. Phase 1 — Security Hardening (C1–C5)

Issue	File	Root Cause	Fix Applied
C1	PortalController	Plaintext password comparison	BCrypt.Verify() enforced; plaintext path removed entirely
C2	ApiController	No API authentication on /api/events/ingest	Session check + X-Device-API-Key header with constant-time compare
C3	ApiController	Unrestricted EventType field	Allowlist {detection, merge, conflict, speeding, fault}; 400 on violation
C4	All mutating controllers	Missing ModelState.IsValid checks	Guard added at top of every POST action across 6 controllers
C5	AppDbContext / Program	EnsureCreated() in startup creating DDL	MigrateAsync() used for PostgreSQL; EnsureCreated() only for SQLite dev

3.1 SessionAuthFilter

A global MVC action filter checks session for UserId on every request. The new [SkipSessionAuth] attribute exempts public endpoints (e.g. /health, /Portal/Login). The filter short-circuits to 401 JSON for API requests and 302 redirect to /Portal/Login for browser requests.

4. Phase 2 — Architecture Fixes (A1–A8)

Issue	File	Root Cause	Fix Applied
A1	TrafficService + others	static new Random() — not thread-safe	Random.Shared throughout; BuildSimulatedSegments patched in P7
A3	VehicleEventsController	Public feed exposed unauthenticated	Session check added; feed returns 401 if no session
A4	ConfigService	TomTom key read only from appsettings	Environment variable TOMTOM_API_KEY takes priority
A5	VehicleEventsController	Duplicate controller existed	Merged into single controller; duplicate removed
A6	DatabaseViewerController	Raw password visible in DB viewer	Password field masked as *** in all DB viewer outputs
A7	TriangulationController	POST actions returned full page reload	AJAX responses for all mutation actions
A8	DashboardController	IDOR — any highway ID accepted in session	Session HighwayId validated against user-owned highways

5. Phase 3 — Design Improvements (D1–D7)

Issue	File	Root Cause	Fix Applied
D1	Models/Enums.cs + AppDbContext	Magic strings for status/type fields	C# enum definitions + EF Core HasConversion() string converters
D4	Data/DbInitializer.cs	Seed data lacked altitude values	Realistic altitude ranges seeded on all switch servers and sensors
D5	Services/VehicleRegistry.cs	Static class, untestable	IVehicleRegistry interface + DI singleton; injected into VehiclesController
D6	Services/HeartbeatMonitorService.cs	No device liveness tracking	IHostedService polls every 2 min; marks stale devices offline
D7	appsettings.json	AllowedHosts: "*" in production	Documented with override guidance; Production appsettings recommended

6. Phase 4 — Altitude & 3D Air Scene

6.1 Model Changes

Four models received `AltitudeMeters` (double?) columns. SwitchServer received three additional altitude-band columns:

Model	New Columns	Default / Constraint
SwitchServer	AltitudeMinMeters, AltitudeMaxMeters, AltitudeWidthMeters	double?, nullable — all three required for band rendering
VehicleEvent	AltitudeMeters	double?, null = ground-level event
SensorDevice	AltitudeMeters	double?, null = ground-mounted device
MergeZone	AltitudeMeters	double?, null = surface-level zone

6.2 3D Air Scene — AirScene.cshtml

New route `/Traffic3D/AirScene`. Three-panel layout sharing one Three.js scene graph. Only the camera differs per panel:

Panel	Camera Type	Axis / Angle	Primary Use
A — Perspective	PerspectiveCamera FOV 55°	Orbit (drag + scroll)	Full 3D overview; altitude + lateral spread
B — Side / Elevation	OrthographicCamera	+X axis looking to origin	Altitude stacking; band membership visible on Y
C — Top / Plan	OrthographicCamera	-Y axis (straight down)	Lane spread; corridor separation; XZ footprint

6.3 Altitude Scale

1 metre = 0.05 Three.js scene units (ALT_SCALE) 10 m → 0.50 units (vertiport pad floor) 150 m → 7.50 units (urban corridor ceiling) 800 m → 40.00 units (express corridor ceiling)

6.4 Altitude Band Seed Data

DbInitializer seeds switch servers with two corridor types:

Corridor Type	AltitudeMin (m)	AltitudeMax (m)	Width (m)	Example Zone
Urban Air Mobility	30	150	30	I20 Dallas West / Grand Prairie
Express / Inter-city	200	400	60	I35 Georgetown / I10 Houston
Vertiport Approach	0	50	20	I45 Houston North

7. Phase 5 — Production Enhancements (E1–E6)

E2 — Audit Trail

AuditLog model — append-only, never updated after insert. Fields: UserId, FullName, HighwayId, Controller, Action, EntityType, EntityId, Summary, IpAddress, CreatedDate. Three indexes: UserId, CreatedDate, (HighwayId, CreatedDate).

AuditService — scoped DI service, reads session context, writes AuditLog on every mutating action. Non-fatal: exceptions are swallowed so audit never breaks primary operations.

E3 — Rate Limiting

Policy	Route	Limit	Window	Keyed By
LoginPolicy	POST /Portal/Login	10 requests	15 minutes	Client IP
IngestPolicy	POST /api/events/ingest	60 requests	1 minute	Client IP
ApiReadPolicy	GET /api/*	120 requests	1 minute	Client IP

E4 — Health Endpoint

GET /health — exempted from SessionAuthFilter via [SkipSessionAuth]. Returns 200 with DB ping, entity counts, and runtime/uptime info. Returns 503 if DB is unreachable.

E5/D6 — Heartbeat Monitor

HeartbeatMonitorService (IHostedService) polls every 2 minutes. SensorDevices with LastHeartbeat older than 5 minutes are marked offline. SwitchServers with LastHeartbeat older than 10 minutes are marked degraded. Configurable via appsettings HeartbeatMonitor section.

E6 — Serilog Structured Logging

LoggingExtensions.AddSerilogLogging(): console sink (compact JSON in prod, pretty in dev), rolling daily file at /logs/mss-YYYYMMDD.log with 7-day retention, optional Seq sink via SEQ_URL environment variable. HTTP access log via UseSerilogRequestLogging with custom log level rules (5xx=Error, 4xx=Warning, 2xx=Information).

8. Phase 6 — Vehicle Classification Engine

8.1 VehicleClassifier — 4-Step Algorithm

STEP 1b (Phase 8 – AirFlyCar gate, highest priority) sourceType == "airflycar" flight_phase ∈ {ground, boarding, deboarding, charging} → air_vertiport altitude_m ≤ 5 → air_vertiport altitude_m ≤ 150 → air_urban altitude_m > 150 → air_express (no altitude, speed > 60 mph) → air_urban STEP 1 (Standard altitude gate) altitude_m > 10 → Domain = AIR ≤ 150 m → air_urban > 150 m → air_express STEP 2 (Explicit vehicle_type field – ground only) sedan | suv | truck | motorcycle | van → Category = that value STEP 3 (Speed heuristic – ground, no vehicle_type) speed_mph ≤ 25 → motorcycle speed_mph ≤ 55 → sedan speed_mph ≤ 75 → suv speed_mph ≤ 90 → truck speed_mph > 90 → van STEP 4 (Confidence score – 0.0 to 1.0) vehicle_type present +0.25 altitude present +0.25 source weight +0.12-0.22 speed present +0.10 lat/lon present +0.10 flight_phase present +0.08 < 0.40 → flagged as low_confidence

8.2 Source Type Trust Weights

Source Type	Weight	Rationale
airflycar	0.22	Dedicated UAM telemetry — highest fidelity
physical	0.20	On-site camera/LiDAR/radar — direct observation
satellite	0.18	GPS positioning — high accuracy, moderate latency
tracker	0.16	RFID tag reader — ID confirmed, position estimated
telecom	0.14	Cell network triangulation — lowest spatial accuracy

8.3 Geometry per Category (Three.js)

Category	Domain	3D Geometry	Top View	Colour
sedan	ground	BoxGeometry 1.6×0.4×0.8	Filled rectangle	#3b82f6 Blue
suv	ground	BoxGeometry 1.9×0.55×0.9	Filled rectangle	#22c55e Green
truck	ground	BoxGeometry 2.4×0.75×0.95	Filled rectangle	#f59e0b Amber
motorcycle	ground	CylinderGeometry r=0.15	Small circle	#a855f7 Purple
van	ground	BoxGeometry 2.1×1.0×1.0	Filled rectangle	#64748b Slate
air_urban	air	ConeGeometry r=0.55 (inv)	Circle (wingspan Ø)	#00bcd4 Cyan
air_express	air	OctahedronGeometry r=0.65	Diamond (rotated sq)	#ff6b35 Orange-red
air_vertiport	air	TorusGeometry r=2.5	Concentric rings	#f0abfc Pink

8.4 Database Columns Added

Column	Table	Type	Purpose
VehicleMode	VehicleEvents	TEXT(10) default "ground"	Fast domain filter (ground air)

Column	Table	Type	Purpose
VehicleCategory	VehicleEvents	TEXT(20) default "sedan"	Category for geometry selection
VehicleClassJson	VehicleEvents	TEXT(800) nullable	Full VehicleClass JSON blob

9. Phase 7 — Classification Wiring & Scene Completion

9.1 InputPayloadService Fixes

- **A1 Random.Shared** — static readonly Random _rng = new() replaced with Random.Shared everywhere — thread-safe, no lock needed.
- **P6a Altitude domain-awareness** — altitude_m now generated in domain-consistent ranges: ground 0–5 m (sensor noise), air_urban 30–149 m, air_express 151–800 m.
- **P6b Air vehicle_type** — For satellite and tracker source types, 30% probability of generating air_urban or air_express vehicle_type, making cross-domain simulation realistic.

9.2 Traffic3DController

- A1 — BuildSimulatedSegments patched to use Random.Shared (no more per-call new Random())
- Both Index and AirScene actions query last 80–120 VehicleEvents with VehicleMode + VehicleCategory + VehicleClassJson projected
- Pre-classified data seeds the Three.js scene on first load — no extra API call needed from JS
- CategoryBreakdown dictionary passed to view for legend counts

9.3 Classification Result Panel (DataInputFormats Simulation Tab)

After every SimulationPost or GeneratePayloadAjax call, a result card appears showing:

- Vehicle icon (■■■■■) + category name in uppercase
- Domain badge — GROUND (blue) or AIR (amber)
- Confidence progress bar — green ≥65%, yellow ≥40%, red <40% with ■ warning
- Speed, altitude, source type fields
- Colour-matched category badge matching Three.js render colour

10. Phase 8 — AirFlyCar Simulation & Test Data

10.1 Test InputFormatConfigs — 4 Seeded Records

Config Name	SourceId	Input Source	Key Fields
UAM Telemetry Stream	AFC-UAM-001	wss://telemetry.uam-hub.local/stream	altitude_m, rotor_rpm, battery_soc, flight_phase, corridor_id, pilot_id
Vertiport ADS-B Feed	AFC-ADSB-002	udp://vertiport-receiver-01:30003	icao_address, squawk, nic, nac_p, vertical_rate_fpm, heading
UTM Corridor Position	AFC-UTM-003	https://utm-gateway.local/api/v2/positions	corridor_deviation_m, conflict_flag, separation_m, 4D trajectory
eVTOL Fleet Tracker	AFC-FLT-004	https://fleet.evtol-ops.local/api/telemetry	rotor_health, motor_temp_c, noise_db, passenger_count, destination_pad

10.2 GenerateAirFlyCar() — Correlated Field Generation

The generator is flight_phase-driven. Selecting a phase causes all correlated fields to be generated in physically consistent ranges:

flight_phase	altitude_m range	speed_mph	vertical_rate_fpm	Notes
cruise	30–800 m	100–180	–50 to +50	Stable flight; battery drains steadily
climb	30–800 m	40–120	+300 to +1200	High rotor RPM; motor temp elevated
descent	30–800 m	30–100	–200 to –800	Regenerative if supported
hover	30–150 m	0–15	–20 to +20	Highest noise_db; high rotor RPM
approach	30–150 m	20–70	–50 to –400	event_type="merge"; conflict possible
boarding	0–3 m	0	0	Category=air_vertiport; passengers loading
charging	0–3 m	0	0	Category=air_vertiport; low battery SOC
ground	0–3 m	0	0	Category=air_vertiport; parked on pad

10.3 Conflict Detection

8% probability of conflict_flag=true per generated record (non-ground phases only). When conflict is detected: event_type overridden to "conflict", separation_m set to 50–300 m (vs 400–2000 m normal), VehicleClass.Color overridden to #ef4444 (red), and a pulsing red TorusGeometry ring is rendered around the craft in all three scene panels.

10.4 AirScene.cshtml — View-Specific Geometry

Category	Perspective Panel	Top / Plan Panel	Side / Elevation Panel
air_vertiport	TorusGeometry r=2.5 (ring)	Two concentric RingGeometry	RingGeometry (flat)
air_urban	ConeGeometry r=0.55 (inverted)	CircleGeometry wingspan Ø	CircleGeometry r=0.5
air_express	OctahedronGeometry r=0.65	CircleGeometry ×4 rotated 45° (diamond)	CircleGeometry r=0.5
sedan/suv/van	BoxGeometry (proportional)	PlaneGeometry (footprint)	PlaneGeometry (side profile)
motorcycle	CylinderGeometry r=0.15	CircleGeometry r=0.25	PlaneGeometry 0.6×0.5
Conflict craft	Above + red override	Pulsing TorusGeometry ring overlay	Pulsing ring overlay

10.5 Altitude Ruler

A Canvas 2D ruler is drawn on the right edge of the side panel. Marks at 0, 5, 30, 50, 100, 150, 300, 500, 800 m with colour-coded band thresholds. Automatically scaled to canvas height.

11. Data Model

11.1 Entity Relationship Summary

Entity	Key Fields	Relationships
Highway	HighwayId (PK), Name, State, IsActive	1:N → MergeZone, SwitchServer, SensorDevice, VehicleEvent, UserProfile
MergeZone	ZoneId (PK), HighwayId (FK), Status, AltitudeMeters	1:N → SwitchServer, SensorDevice, TriangulationConfig
SwitchServer	ServerId (PK), ZoneId (FK), AltMin/Max/WidthMeters, Status	N:1 → MergeZone; provides altitude band definition
SensorDevice	DeviceId (PK), ZoneId (FK), AltitudeMeters, LastHeartbeat	N:1 → MergeZone; monitored by HeartbeatMonitorService
VehicleEvent	Id (PK), HighwayId, ZoneId, AltitudeMeters, VehicleMode, VehicleCategory, VehicleClassJson	N:1 → Highway; classified by VehicleClassifier on write
InputFormatConfig	Id (PK), SourceType (physical satellite telecom tracker airflycar), EnabledFieldsRaw	Source configuration for payload generation
SamplePayload	Id (PK), ConfigId (FK), SourceType, Payload (JSON)	N:1 → InputFormatConfig; generated by InputPayloadService
AuditLog	Id (PK), UserId, HighwayId, Controller, Action, CreatedDate	Append-only audit; written by AuditService
UserProfile	Id (PK), HighwayId (FK), UserType, PasswordHash	Session-based auth; BCrypt hashed passwords

12. API Reference

Method	Route	Auth	Rate Limit	Description
GET	/health	None ([SkipSessionAuth])	—	DB ping + entity counts + uptime. 200 OK or 503.
GET	/api/stats	Session or API key	120/min	Zone/server/sensor/event counts; ground+air split.
GET	/api/zones	Session or API key	120/min	All MergeZones for highwayId. Includes AltitudeMeters.
GET	/api/events/live	Session or API key	120/min	Last N events. Params: highwayId, take, mode, category.
GET	/api/altitudebands	Session or API key	120/min	SwitchServer altitude bands for a highway.
POST	/api/events/ingest	Session or API key	60/min	Ingest single event. Body: IngestPayload. Classifies on arrival.
POST	/Portal/Login	None	10/15min	Session login. Accepts UserId + Password.
POST	/DataInputFormats/SimulationPost	Session	—	Generate + classify + store payload. Returns classification JSON.

12.1 GET /api/events/live — Response Shape

```
[ { "id": 42, "vehicleId": "AFC-7821", "eventType": "detection", "zoneId": "I20-Z001", "highwayId": "I20-TX", "speedMph": 135.0, "latitude": 32.7812, "longitude": -96.9934, "altitudeMeters": 87.5, "vehicleMode": "air", "vehicleCategory": "air_urban", "vehicleClassJson": "{ \"domain\": \"air\", \"category\": \"air_urban\", \"color\": \"#00bcd4\", ... }", "createdDate": "2026-06-19T16:42:00Z" } ]
```


13. Security Architecture

13.1 HTTP Security Headers

Header	Value	Purpose
X-Frame-Options	DENY	Clickjacking prevention
X-Content-Type-Options	nosniff	MIME-type sniffing prevention
Referrer-Policy	strict-origin-when-cross-origin	Referrer leakage control
Content-Security-Policy	default-src 'self'; script-src 'self' cdn.jsdelivr.net ...	XSS mitigation; CDN allowlist only
Server	(removed)	Technology fingerprinting prevention
X-Powered-By	(removed)	Technology fingerprinting prevention
HSTS	max-age set via UseHsts()	HTTPS enforcement in production

13.2 Session Security

- Cookie: HttpOnly=true, SecurePolicy=Always, SameSite=Strict, Name=__mss
- IdleTimeout: 2 hours
- Password storage: BCrypt with work factor (BCrypt.Net-Next 4.0.3)
- No plaintext password paths — all comparison via BCrypt.Verify()
- Constant-time API key comparison to prevent timing attacks

13.3 Input Validation

- ModelState.IsValid checked on every POST action across 6 controllers
- EventType validated against allowlist {detection, merge, conflict, speeding, fault}
- DuplicateConfig targetTab validated against allowlist of 5 valid values
- bbox parameter validated against regex before use in TomTom API URL
- All string fields have [MaxLength] attributes enforced by EF Core
- RequestSizeLimit(32768) on /api/events/ingest

14. Deployment

14.1 Environment Variables

Variable	Required	Description
DATABASE_URL	Prod only	PostgreSQL connection string (postgres://user:pass@host/db). Parsed at startup; triggers MigrateAsync().
DEVICE_API_KEY	Recommended	API key for device ingest endpoint. Constant-time compared with X-Device-API-Key header.
TOMTOM_API_KEY	Optional	TomTom Traffic Flow API key. Falls back to simulated segments if absent.
SEQ_URL	Optional	Seq log aggregation server URL (e.g. http://seq:5341). Enables Seq sink in Serilog.
PORT	Optional	Kestrel listen port. Defaults to 5000.

14.2 Startup Sequence

```
1. Serilog bootstrap logger initialised (catches pre-DI errors) 2.
WebApplication.CreateBuilder(args) 3. AddSerilogLogging() – full logging pipeline 4. DATABASE_URL
present? → UseNpgsql : UseSqlite 5. DI registrations: MVC, Session, OutputCache, RateLimiter,
AuditService, VehicleClassifier, IVehicleRegistry, HeartbeatMonitorService 6. app.Build() 7.
Scoped: db.Database.MigrateAsync() [PostgreSQL] db.Database.EnsureCreated() [SQLite] 8.
DbInitializer.Seed(db) – idempotent (checks Highways.Any()) 9. Middleware pipeline built and
app.Run()
```

15. Complete File Inventory

File	Phase	Description
Controllers/PortalController.cs	P1/C1	BCrypt-only login; session establishment
Controllers/ApiController.cs	P1-P6	REST API; auth; rate limiting; classify on ingest; live feed
Controllers/MergeZonesController.cs	P1/C4	CRUD + ModelState guards
Controllers/SwitchServersController.cs	P1/C4	CRUD + ModelState + altitude band binding
Controllers/SensorsController.cs	P1/C4	CRUD + ModelState
Controllers/TriangulationController.cs	P1/A7	CRUD + AJAX responses
Controllers/DashboardController.cs	P2/A8	Highway IDOR guard
Controllers/VehicleEventsController.cs	P2/A3,A5	Feed auth + merged controller
Controllers/DatabaseViewerController.cs	P2/A6	Password masking in viewer
Controllers/VehiclesController.cs	P3/D5	DI IVehicleRegistry injection
Controllers/UsersController.cs	P3/D1	Enum normalisation + ModelState
Controllers/Traffic3DController.cs	P4,P7	AirScene action + RecentEvents + Random.Shared
Controllers/DataInputFormatsController.cs	P6,P8	Classify on SimulationPost; airflycar tab
Controllers/HealthController.cs	P5/E4	GET /health endpoint
Services/TrafficService.cs	P2/A1	Random.Shared
Services/ConfigService.cs	P2/A4	Env var priority for TomTom key
Services/VehicleRegistry.cs	P3/D5	IVehicleRegistry singleton
Services/InputPayloadService.cs	P7,P8	Random.Shared; altitude-aware; GenerateAirFlyCar()
Services/VehicleClassifier.cs	P6,P8	4-step classification; airflycar gate; air_vertiport
Services/AuditService.cs	P5/E2	Append-only audit writer
Services/HeartbeatMonitorService.cs	P5/E5	IHostedService device liveness monitor
Models/Enums.cs	P3/D1	Enum definitions for all string-typed fields
Models/SwitchServer.cs	P4	AltitudeMinMeters, AltitudeMaxMeters, AltitudeWidthMeters
Models/VehicleEvent.cs	P4,P6	AltitudeMeters + VehicleMode/Category/ClassJson
Models/SensorDevice.cs	P4	AltitudeMeters
Models/MergeZone.cs	P4	AltitudeMeters
Models/AuditLog.cs	P5/E2	Append-only audit entity
Data/AppDbContext.cs	P3,P5	EF converters + AuditLogs DbSet + indexes
Data/DbInitializer.cs	P3/D4	Altitude-band seed data

File	Phase	Description
Migrations/20260620000000_AddAltitudeFields.cs	P4	Altitude columns on 4 tables
Migrations/20260620000001_AddAuditLog.cs	P5/E2	AuditLogs table
Migrations/20260620000002_AddVehicleClassification.cs	P6	VehicleMode/Category/ClassJson
Migrations/20260619000000_AddAirFlyCarSourceType.cs	P8	4 airflycar InputFormatConfigs seeded
Infrastructure/RateLimiterExtensions.cs	P5/E3	Login/Ingest/API rate limiting policies
Infrastructure/LoggingExtensions.cs	P5/E6	Serilog configuration
Filters/SessionAuthFilter.cs	P1,P5	Global auth filter; [SkipSessionAuth] aware
Filters/SkipSessionAuthAttribute.cs	P5/E4	Public endpoint marker
Views/Traffic3D/Index.cshtml	P6	4-view toggle; domain filter; classification inspector
Views/Traffic3D/AirScene.cshtml	P4,P7,P8	Split/side/top panels; airflycar geometry; alt ruler; conflict rings
Views/DataInputFormats/Index.cshtml	P7,P8	Classification result panel; airflycar tab; eVTOL source option
Views/Shared/_Layout.cshtml	P4	Air Scene nav link
ViewModels/Traffic3DViewModel.cs	P7	RecentEventsJson + ground/air counts + CategoryBreakdown
ViewModels/DataInputFormatsViewModel.cs	P8	AirFlyCarConfigs list
Program.cs	P1-P8	Full pipeline: rate limiter, audit, Serilog, VehicleClassifier, heartbeat
appsettings.json	P5/D7	HeartbeatMonitor + Serilog config keys; AllowedHosts guidance
AirwaysMergeSafeServer.csproj	P5	Serilog package references
VEHICLE_CLASSIFICATION.md	P6	Algorithm design doc: decision tree, geometry table, data flow
AirwaysMergeSafeServer_HLD.pdf	P8	This document — all phases, design decisions, API reference